

Team Delivery Package: EDA Engine

Enterprise Milestone-Based Technical Specifications

MODULE NAME:	Team 3: EDA Engine
GIT BRANCH:	feature/eda-visualization
LEAD MEMBERS:	Mohamed Kamal, Yomna Ashraf, Samar Mahmoud
WORKLOAD TIME:	3 weeks
KICKOFF DATE:	11 July 2026
CODE FREEZE:	23 July 2026
VERSION:	v1.2.0

1. Cover Page & Metadata Summary

This delivery package establishes the contract specifications for the **Team 3: EDA Engine**.

Milestone	Target Date	Status	Description
Project Kickoff	11 July 2026	COMPLETED	Milestone scope locked & repository structured.
Code Freeze	23 July 2026	IN PROGRESS	All functional packages and tests locked.
Integration Phase	24 July 2026	PLANNED	Multi-team pipeline joins and bug resolution.
Final Presentation	25 July 2026	PLANNED	Final delivery and platform presentation.

2. Team Overview & Assignments

Resource Details

Lead Team Members: **Mohamed Kamal, Yomna Ashraf, Samar Mahmoud**

Branch Space: ``feature/eda-visualization``

Role Assignment: **Exploratory Data Analysts & Visual Engineers**

Difficulty Rating: **Medium** | Priority Index: **High** | Est. Workload: **3 weeks**

3. Business & Technical Goals

Business Objective

Provide visual summaries and correlation indices to uncover hidden patterns in numeric data.

Technical Objective

Calculate descriptive statistics transposes, correlation matrices, and structure metadata for Plotly widgets.

Position inside AutoAnalyst AI

Post-ingestion analysis layer (third step in the pipeline).

4. System Architecture & folder layout

Assigned folder Tree Structure

```
src/autoanalyst/  
  eda/  
    __init__.py  
    analyzer.py  
  reports/  
    figures/
```

Target Code Modules

Your codebase modifications belong inside: `src/autoanalyst/eda/analyst.py`.

Functional Code Interface Specifications

```
# src/autoanalyst/eda/analyst.py
import pandas as pd

def get_numeric_summary(df: pd.DataFrame) -> pd.DataFrame:
    """Generates descriptive statistics for all numeric columns (describe transpose)."""
    pass

def get_correlation_matrix(df: pd.DataFrame, method: str = "pearson") -> pd.DataFrame:
    """Generates a correlation matrix for numeric features."""
    pass
```

API Specification

```
get_numeric_summary(df: pd.DataFrame) -> pd.DataFrame
Returns:
- DataFrame with indices representing column names, and metrics (count, mean, std, min, 25%, 50%, 75%, max)
  as columns.
```

5. Git Workflow & Code Integration Policy

Branching Workflow

All developers pull and write inside `feature/eda-visualization` created from the latest `develop` branch.

Pull Request Requirements

1. Merge latest develop branch changes into features before raising PRs.
2. Assign team lead reviews and resolve conflicts.
3. A minimum of one approved review check is required.

6. Quality Standards & Guidelines

Format & Lint Rules

All variables, methods, and file layouts must use lower_case snake_case parameters. Code comments must be in English.

Error Checking & Imputers

All functional blocks should be guarded by custom exception checks (raise ValueError/KeyError).

Logging Rules

Incorporate logging warnings rather than std print statements.

7. Expected Deliverables & Verification

Required Files list

`src/autoanalyst/eda/analyzer.py`

Verification Code Assertions

Unit test validations are situated in: `tests/test\_eda.py`.

Pipeline integration assertions belong in: `tests/test\_pipeline.py`.

IMPORTANT GUIDELINE

All functional checks must compile successfully with zero syntax warnings before code freeze.

8. Definition of Done Checklist

- [] All functions compile cleanly with PEP 8 checks.
- [] Unit tests cover all key methods and functions.
- [] Try-except checks isolate external dataframe issues safely.
- [] The PR has been reviewed and verified by Team 1.

9. Common Mistakes & Troubleshooting

Pitfalls to Avoid

- Failing to check if numeric columns exist before calling describe().
- Calculating correlation matrices when less than 2 numeric columns are present.
- Rendering massive scatter matrix graphs on datasets with 100+ columns.